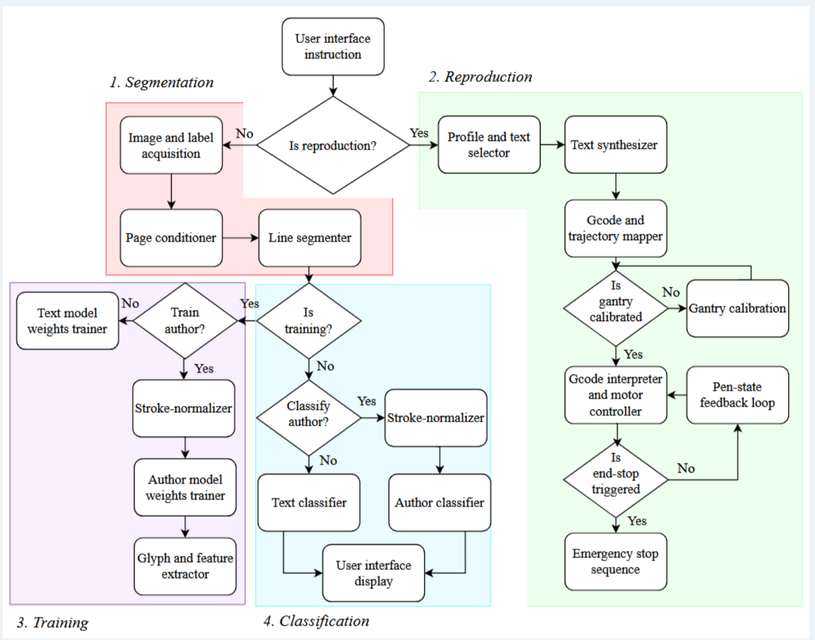
Handwriting Robot

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Project overview

The goal is to recognise handwriting and reproduce new text while retaining an individual's writing style. A photographed page is corrected and split into lines. Separate neural networks recognise its text and writer. Labelled samples provide glyph shapes and spacing for stored writer profiles. New text is assembled from these profiles and converted into G-code. An ODROID N2+ controls the gantry and pen to draw it on paper.

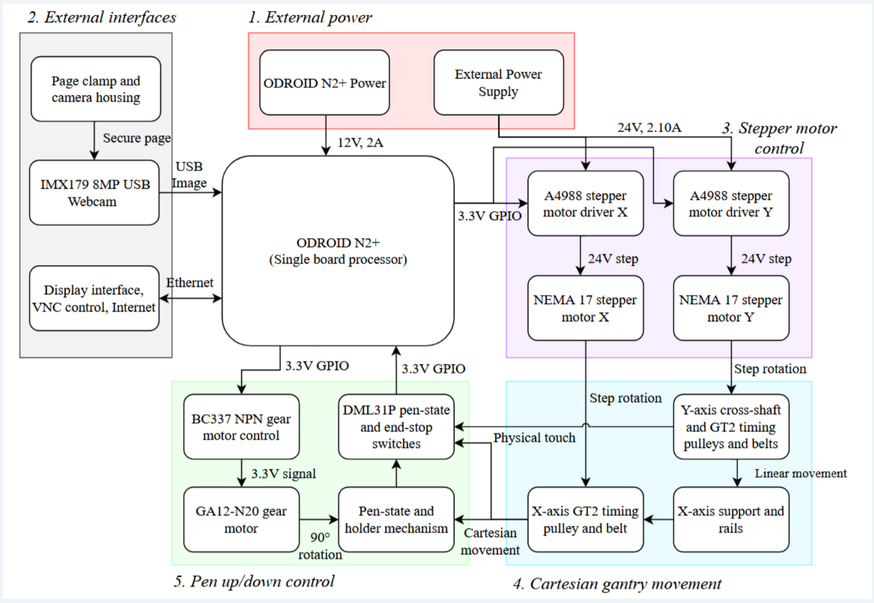
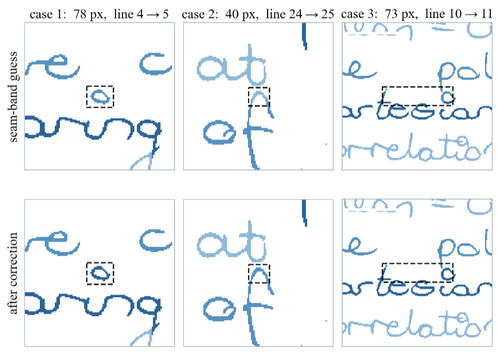
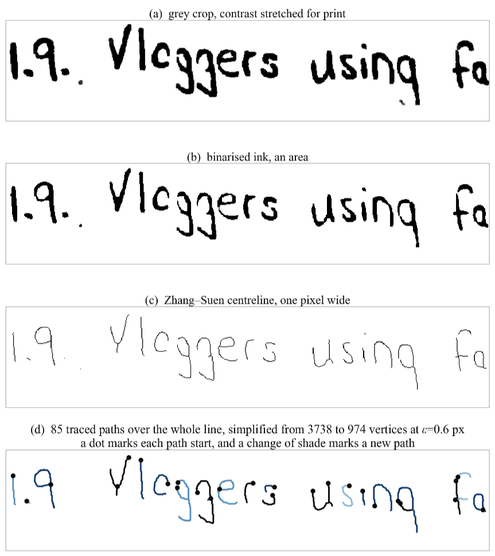


Image processing

Local background normalisation reduces shadows. Otsu's thresholding extracts the ink, and projection profiles with dynamic programming seams separate the lines. Zhang and Suen thinning recovers stroke centrelines.

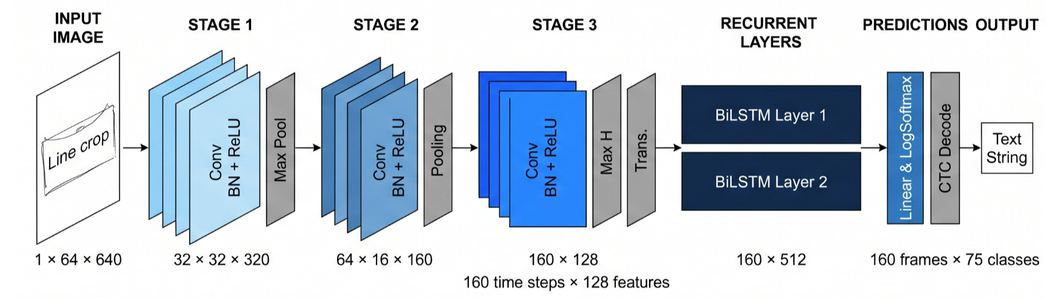




Recognition

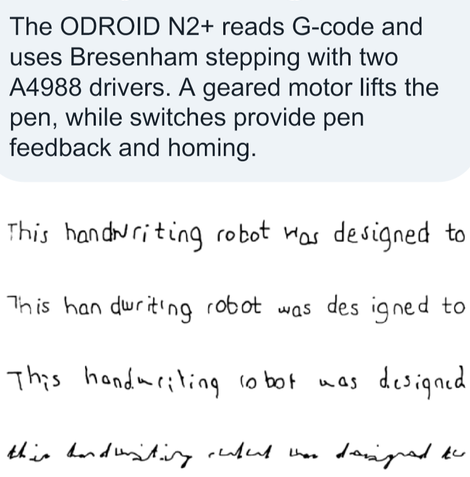
A CNN extracts line features. Two BiLSTM layers provide sequence context, and CTC decodes the text. A separate CNN identifies the writer from strokes normalised to 0.17 x-height.

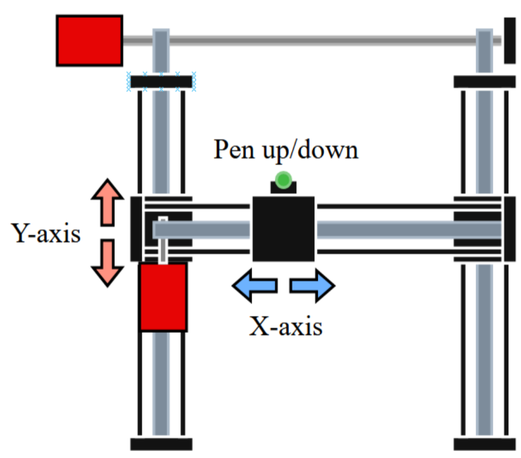
Parameter	Setting
Reader input	64 × 640 px
Writer styles	11
Stroke width	0.17 x-height



Gantry Plotting

The ODROID N2+ reads G-code and uses Bresenham stepping with two A4988 drivers. A geared motor lifts the pen, while switches provide pen feedback and homing.

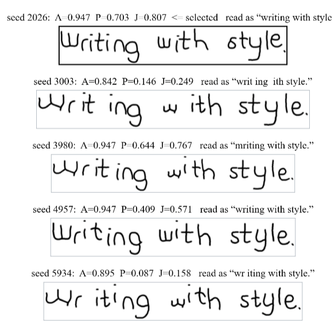


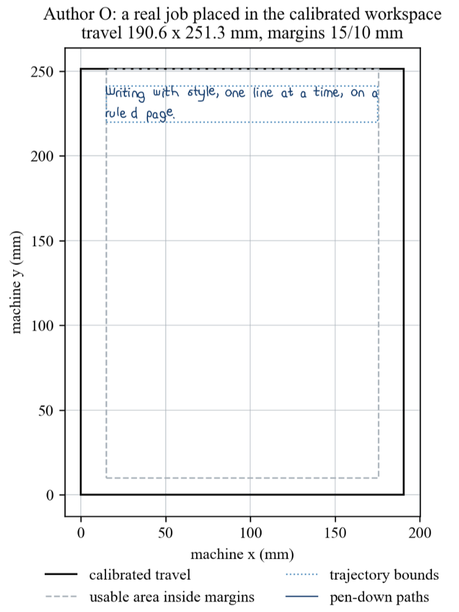


Handwriting synthesis

CTC forced alignment locates glyphs in labelled lines. Skeleton tracing stores their paths in writer profiles. The synthesiser selects variants, sets spacing and joins, then maps the paths to G-code.







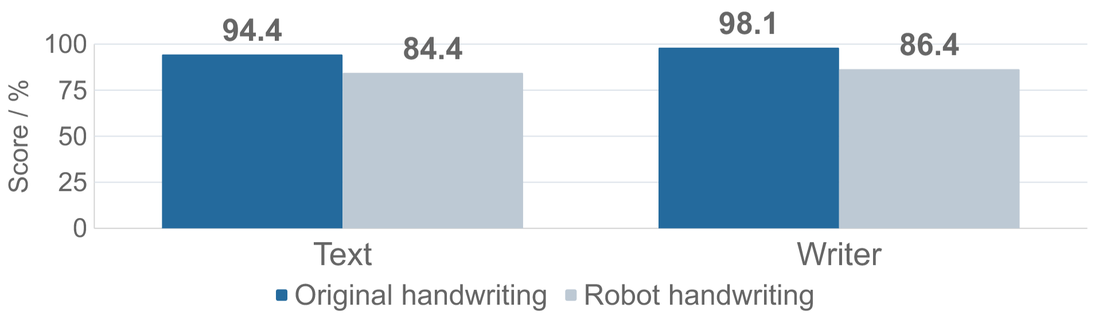
Measured performance

86.7%
relative shape score
against natural handwriting variation

0.297 mm
mean absolute length error

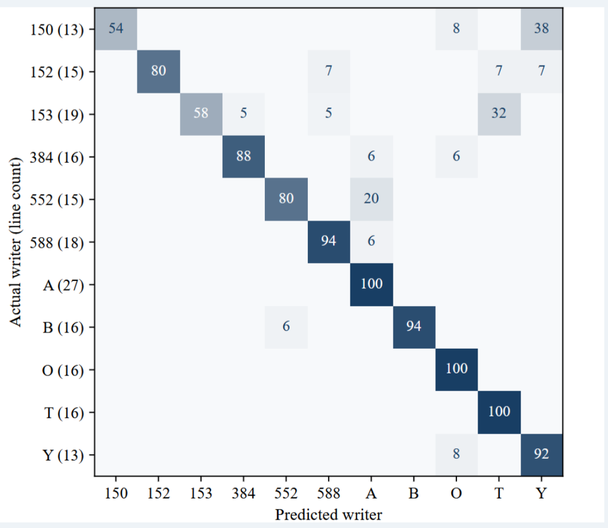
2.43 s
average completion time per character, 25 jobs

Recognition scores



Original validation and robot output are separate test sets.
Classification time remains above target.

Robot writer identification



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